

MARCEL ROTH

in |  |  marcelroth100@aol.com

PROFESSIONAL SUMMARY

Machine Learning Engineer with a proven record of solving novel problems in complex domains. Spearheaded the machine learning effort that recovered the first title from a 2,000-year-old carbonized scroll, a landmark achievement in digital archaeology. Experienced in developing and automating end-to-end ML systems, from data curation, model training, evaluation to deployment. I thrive in fast-paced, competitive environments and excel at spinning up quickly in new technical domains.

PROFESSIONAL EXPERIENCE

denkbare GmbH

Machine Learning Engineer 1 yr 11 mos

Engineered production-grade ML pipelines including fine-tuned multimodal LLMs for Document AI and embedding-based concept search, while building full-stack annotation tools for data curation and benchmarking.

11/2023 – present, Würzburg, GER

Julius-Maximilians-Universität Würzburg

Chair for HCI, Research Intern 6 mos

Developed automated data processing pipelines to standardize large-scale XR motion datasets, curating 15M recordings for cross-dataset analysis and interoperability [1].

10/2023 – 12/2024, Würzburg, GER

Chair for Data Science, Research Assistant 8 mos

Conducted in-depth review of the latest research on ecosystem forecasting models and synthesized key findings into recommendations for use in vegetation modeling research project.

08/2023 – 03/2024, Würzburg, GER

Chair for AI & KS, Research Intern 6 mos

Fine-tuned Foundation Models for few-shot medical image classification (thorax, pathology, endoscopy) using transfer learning and domain adaptation in limited data scenarios.

08/2023 – 02/2024, Würzburg, GER

eXXcellent solutions GmbH

Software Engineer, Working Student 2 yrs 9 mos

Developed LLM-based writing assistant for legal documents, enhanced web-based document editors with custom components, and built full-stack modules for internal HR management applications. 02/2021 – 11/2023, Ulm, GER

SELECTED SKILLS

Programming Languages (by relevance)

Python 3 yrs, Java 4 yrs, TypeScript/JavaScript 4 yrs

Machine Learning & Computer Vision

PyTorch, Lightning, wandb, albumentations, timm, NumPy, Pandas, Pillow, OpenCV, Scikit-Learn

Web & Infrastructure

React, Vue.js, PostgreSQL, SQLite, Docker, slurm

Languages

German, English C1, Swedish A1

EDUCATION & COURSES

Julius-Maximilians-University Würzburg

M.Sc. Computer Science, Grade: 1.1 with distinction

Thesis: "Systematic Evaluation of Deep Learning Pre-training Strategies for Gastrointestinal Endoscopy".

10/2022 – 09/2025, Würzburg, GER

University of Applied Sciences Ulm

B.Sc. Computer Science, Grade: 1.5

Exchange Semester: Halmstad University, Grade: 1.0

07/2018 – 09/2022, Ulm, GER

AISF Alignment Course

3-month course on AI safety and alignment, covering AI risks, scalable oversight, interpretability and governance.

09/2023 – 11/2023

SELECTED AWARDS

Vesuvius Challenge

1st Title Prize (\$60,000)

Recovered the first known title from a 2,000-year-old carbonized Herculaneum scroll using a custom-built transformer-based segmentation model [2].

2025

Plus three additional progress prizes:

- May 2025 Progress Prize (\$1,000)
- November 2024 Progress Prize (\$1,000)
- Grand Challenge 2023 Progress Prize (\$1,000)

Magnet4Cardiac7T Spring School

1st Place

Optimized ultra-high-field MRI coil configurations, balancing magnetic field homogeneity and tissue heating.

2025

"Tierzählstation" Challenge

1st Place

Developed a deep learning model for wildlife animal population monitoring.

2024

Capsule Vision 2024 Challenge

3rd Place

Developed a deep learning model for medical image multi-class classification in gastrointestinal endoscopy [3].

2024

NeurIPS 2023 MedFM Challenge

2nd Place

Applied few-shot learning for thorax, pathology, and endoscopy classification, improving limited data scenarios.

2023

PUBLICATIONS

- [1] Christian Rack et al. Navigating the Kinematic Maze: Analyzing, Standardizing and Unifying XR Motion Datasets. IEEE. 2024.
- [2] Jo Marchant. AI reveals title of 'unreadable' Vesuvius scroll for first time. Nature. 2025.
- [3] Marcel Roth et al. Domain-Adaptive Pre-training of Self-Supervised Foundation Models for Medical Image Classification in Gastrointestinal Endoscopy. ArXiv. 2024.